Implementation of Data Integrity Constraints

Primary keys, foreign keys, unique, check, default, and NOT NULL rules

Prepared by: Hemanth Krishna S

Project: Hotel Management System. Output convention: result tables are expected outputs from the fixed rows in the shared SQL script; they are not screenshots of MySQL execution.

1. Integrity constraints

Constraints make invalid relationships and values fail close to the data. The hotel schema uses primary keys, foreign keys, unique identifiers, checks, defaults, and NOT NULL declarations.

Constraint	Where used	Purpose
PRIMARY KEY	Every table	Prevents duplicate or missing row identity.
FOREIGN KEY	Rooms, staff, bookings, booking_services, payments	Requires referenced parent records to exist.
UNIQUE	Guest number/email, hotel name+city, room number per hotel, service name	Prevents duplicate business identifiers.
CHECK	Rates, occupancy, dates, status, quantities, payment amount	Rejects values outside the encoded rules.
DEFAULT	Guest registration date, room status, booking status, service quantity, payment status	Provides an initial value when omitted.
NOT NULL	Required identity and descriptive fields	Prevents incomplete records.

2. Successful verification queries

After loading the shared sample data, the following queries should return zero invalid or orphan rows.

```
SELECT COUNT(*) AS orphan_room_rows
FROM rooms r LEFT JOIN hotels h ON h.hotel_id=r.hotel_id
LEFT JOIN room_types rt ON rt.room_type_id=r.room_type_id
WHERE h.hotel_id IS NULL OR rt.room_type_id IS NULL;

SELECT COUNT(*) AS orphan_booking_rows
FROM bookings b LEFT JOIN guests g ON g.guest_id=b.guest_id
LEFT JOIN rooms r ON r.room_id=b.room_id
WHERE g.guest_id IS NULL OR r.room_id IS NULL;

SELECT COUNT(*) AS invalid_rate_rows
FROM room_types WHERE base_rate < 0;

SELECT COUNT(*) AS duplicate_guest_email_groups
FROM (SELECT email FROM guests GROUP BY email HAVING COUNT(*) > 1) d;

SHOW CREATE TABLE guests;
```

Check	Expected result
orphan_room_rows	0
orphan_booking_rows	0
invalid_rate_rows	0
duplicate_guest_email_groups	0
SHOW CREATE TABLE guests	Displays primary key and unique guest number/email constraints.

3. Constraint rejection examples

Run each statement separately only if you want to capture a rejection message. These statements are intentionally invalid and should not be included in the normal seed script.

```
-- Expected duplicate email rejection
INSERT INTO guests (guest_id, guest_no, full_name, email, phone, registered_on)
VALUES (99,'GST099','Duplicate Email Test','ananya.rao@example.com',
        '+91-90000-19999','2026-10-25');

-- Expected CHECK rejection: negative rate
INSERT INTO room_types (room_type_id,type_name,max_occupancy,base_rate)
VALUES (99,'Invalid Rate',2,-100.00);

-- Expected FOREIGN KEY rejection: unknown guest
INSERT INTO bookings (booking_id,guest_id,room_id,check_in,check_out,booking_status)
VALUES (999,999,1,'2026-11-01','2026-11-03','CONFIRMED');
```

4. Demonstrating a default

```
START TRANSACTION;
INSERT INTO bookings (booking_id, guest_id, room_id, check_in, check_out)
VALUES (199,5,8,'2026-12-01','2026-12-03');
SELECT booking_id, booking_status FROM bookings WHERE booking_id=199;
ROLLBACK;
```

Expected result: booking_id 199 has booking_status CONFIRMED. ROLLBACK removes the demonstration row, leaving the original sample data unchanged.